

Inert IR Command Translator

Map one valid copied symbol to a different local symbol

E0 POLICY COMPOSITION — NO RECEIVER, EMITTER, OR DEVICE CONTROL

Lesson 054 — project
Prerequisites Lessons 052 and 053
Evidence distinct named memory cells

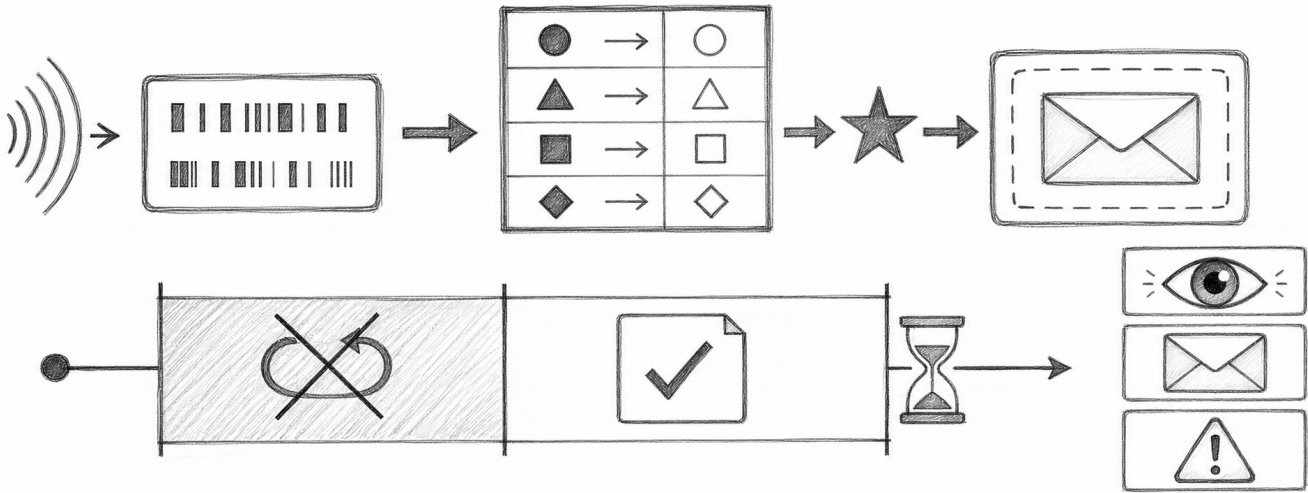
Time 150–210 minutes
Board compile-only Mega replay
Boundary fixed allowlist only

Project boundary. InertIrTranslator composes copied receive evidence with an immutable local emission catalog. It owns no receiver, emitter, pin, timer, interrupt, carrier endpoint, display, bus, supply, or optical path. It is not a learning remote, replay bridge, or unknown-protocol transmitter.

1 Translate through a fixed different-symbol map

Valid received symbol	Local output symbol	Rule
StationPing	StationReady	fixed firmware mapping
StationReady	StationAcknowledge	fixed firmware mapping
StationCancel	none	cancellation dominates
StationAcknowledge	StationPing	fixed firmware mapping

Only KnownValid, IntegrityVerified evidence from the configured qualified source can select a mapping. Every output differs from its input. Repeat, unknown, malformed, truncated, overflowed, stale, source-faulted, self-echo, and unlisted records remain visible but cannot become transmit authority.



Alternative text: graphite copied marks enter a fixed different-symbol table and then a bounded envelope card. Below, a crossed self-echo interval precedes a checked response window and hourglass boundary; eye, envelope, and warning result cells remain separate.

2 Admit one complete atomic envelope

IrTranslatorUpdateInput carries cancellation, a prepared commit, receive evidence, and actual-emission evidence at one supplied time. Validate the complete envelope before mutation, then apply fixed precedence:

1. reject a structurally invalid envelope without mutation;
2. preserve an already latched dependency or safety fault;
3. apply matching cancellation and invalidate both candidates;
4. copy and classify receive evidence;
5. correlate actual-emission evidence;
6. commit a still-current parent and child candidate; and
7. publish presentation intent last.

There are no separate receive, cancel, commit, or emission entry points whose caller-selected order could change the result.

3 Bind parent and child capabilities

`prepareTranslation(operationId, now)` uses only the current Lesson 052 evidence generation and fixed map. Its preview binds translator owner, epoch, configuration, mapping digest, parent generation, operation, copied input digest, receive provenance, evidence generation, different output symbol, catalog identity, and the complete Lesson 053 child preview.

A foreign, stale, reconstructed, reused, reset-invalidated, or shutdown-invalidated preview rejects. A simultaneous valid cancellation invalidates parent and child before either commits. After complete preflight, the child commit and parent publication describe one attributable semantic operation, not optical emission.

4 Separate logical intent from actual emission evidence

A policy commit is not optical start. Round-trip correlation begins only when qualified `IrEmitterEvidence` supplies the exact source, transaction, actual start, actual completion, and status. E0 supplies explicitly synthetic values.

Half-open interval	Classification
$[actual\ start, actual\ completion + guard)$	SelfEchoSuppressed ; visible, never a response
$[response\ start, response\ start + window)$	eligible only after every attribution check
exact response end	timeout; exclusive boundary

All interval arithmetic uses unsigned modular half-range comparison. Overflow, regression, and exact-half-range ambiguity reject or fault atomically. Lesson 052 supplies observation time, not optical arrival time, so E0 proves only deterministic observation-time correlation.

5 Publish one complete round-trip record

A successful `IrRoundTripResult` atomically binds operation and output symbol, catalog identity, transmitter source, actual start and completion, elapsed duration, receive disposition and strength, complete receive provenance and generation, and correlation status.

Payload equality alone is insufficient. Source, mapping, catalog, operation, transaction, emission evidence, expected response symbol, self-echo exclusion, and response window must all agree. Fields from different updates cannot be assembled into a success.

6 Keep result cells independent

Cell	Shows	Never grants
receive	identity, disposition, strength	transmission authority
translation	fixed different-symbol candidate	electrical emission
transmit	envelope intent and attribution	optical success
self-echo	bounded suppressed count	valid response
round trip	complete correlated record	hardware qualification
fault	independent presentation intent	authority or cancellation

These memory cells are the E0 non-Serial observation path. Compilation does not prove that a powered Mega executed the sketch.

7 Run the deterministic project workbook

1. Initialize both children and the translator. Predict inactive intent, empty round trip, and distinct receive, transmit, and fault cells.
2. Admit each valid mapping and prove every output differs from its input.
3. Try repeat, unknown, malformed, stale, unlisted, and wrong-source evidence. Verify visibility without a translation candidate.
4. Prepare a translation, then try foreign, stale, changed, consumed, reset-invalidated, and shutdown-invalidated previews.
5. Collide valid cancellation with commit, receive, emission evidence, and presentation failure. Verify cancellation precedence.
6. Replay actual-emission evidence and receive at one tick before, exactly at, and one tick after both the guard and response boundaries.
7. Exercise rollover, regression, half-range ambiguity, timeout, missed service, and suppressed-count saturation.
8. Replay two instances with identical fixture inputs and compare every result field byte-for-byte.

frame	receive	strength	output	envelope	round trip
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____

8 Separate acquisition evidence from safe-state evidence

Acquisition prediction and observation. Valid child configuration, fixed storage, nonzero epochs, and immutable map/catalog identities allow E0 initialization. Host tests inspect lifecycle, bindings, generations, digests, and distinct result cells in memory.

Acquisition interpretation. Success proves bounded policy composition only. It does not prove receiver sampling, carrier generation, emitted light, a response, or device control.

Safe-state prediction and observation. Construction, cancellation, completion, fault, shutdown, and restart publish inactive envelope intent and no pending translation. This is not emergency stop, power removal, or proof of a dark emitter.

9 Read the conservative resource evidence

The final canonical Mega 2560 replay measures 16,162 bytes of flash and 1,343 bytes of static SRAM. The maximal measured fixture uses 21,864 bytes of flash and 3,531 bytes of static SRAM. `sizeof(InertIrTranslator)` is 407 bytes, and its required caller-owned receive storage is another 400 bytes, for 807 bytes at that ownership boundary.

Conservative compiler evidence estimates 888 bytes on the maximal analyzed stack path. That misses the 800-byte stack target but passes the 1,024-byte hard limit, leaving about 3,773 bytes after the measured static and stack allowances on the Mega 2560.

These are deliberately conservative compiler and link-map results for the exact fixtures and toolchain, not runtime high-water measurements or proof of every interrupt schedule. Any source, toolchain, optimization, library, interrupt, or aggregate-composition change requires remeasurement.

10 Keep powered work outside this lesson

E1 requires separately qualified exact receiver, emitter and driver, carrier endpoint, observation path, board, supply, complete resource budgets, authoritative formal schematic, default-off evidence, and recorded bench acceptance. Current limiting plus bounded duty reduces exposure but does not establish eye safety. Unknown devices, access-control commands, high-power or long-range transmission, and pyrotechnic or launcher control are prohibited.